

SECOND ORDER

ISSUE 11 · SEPTEMBER 2026

AI, Capital & Work

The Firm's Practice and the Worker's Judgment

AI hands a new hire the firm's best practice as finished output on the first day. Whether that output becomes the hire's own judgment is a separate transfer, and no study here shows it happening.

Four output experiments, two withdrawal trials, one worker survey and the Stanford and BLS employment records, read against Issue 03 and its addendum.

FOR DISCUSSION · NOT INVESTMENT ADVICE

Output and judgment track two different boundaries

THE THROUGH-LINE

Knowledge is either written down or learned by doing, and it is kept either by the worker or by the firm. AI moves knowledge between the firm and the worker in both directions. It does not move it from what a worker can state to what a worker can do. Measured output tracks the first boundary and skill formation tracks the second.

WHY NOW

Issue 03 found AI-exposed occupations growing while hiring froze; the addendum found adopting firms still hiring juniors. The open question was what those juniors learn once the AI does the written-down part of the job. Four experiments and two withdrawal trials now answer it.

WHAT YOU'LL LEARN

1. The two boundaries, and why the entry-level findings stop contradicting each other once placed.
2. Where entry employment fell, and how it fits Issue 03 and the addendum.
3. What the AI hands a new hire, on four definitions of new.
4. What the worker keeps when the AI is taken away.
5. Why the people who hold the practice have a say in the model.
6. Which tasks the AI removes, and what that leaves to learn on.
7. What would change our mind.

IF YOU REMEMBER ONE LINE: A junior's output no longer tells you what the junior knows; measure unaided judgment separately, starting with senior review minutes per piece of junior work.

Output transfers on day one; judgment does not

AI moves knowledge across the boundary between a firm and its workers, in both directions. It does not move knowledge across the boundary inside a worker, between what someone can state and what someone can do. Measured performance tracks the first boundary and skill formation tracks the second, so a rise in one beside a fall in the other is two readings.

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2 months = 6

A support worker with two months on the job and the AI matched a worker with six months and no AI, among 5,172 workers at a single software firm.

5.4 points

Students who practiced with a plain AI scored 5.4 points below students who practiced without one, on an exam taken with no AI, among 825 at a Turkish high school.

11%

Employment of 22-to-25-year-olds in the most AI-exposed jobs fell about 11 percent from November 2022 to June 2026 while the same age group in less exposed jobs grew about 10 percent.

EXHIBIT 1
The evidence stops contradicting itself once each finding is placed

Two distinctions: whether knowledge is written down, and whether the worker or the firm holds it

	THE WORKER	THE FIRM
WRITTEN DOWN	Degree knowledge Rules and answers a person can look up Output produced with an AI: support workers, developers, consultants, adult learners Measures output, not what the worker has learned.	Documented process Prior cases, worked solutions, procedures The firm's record used to train the model; teachers' prepared solutions; workers' data permission A record of the work is not all the expertise behind it.
LEARNED BY DOING	Personal judgment Recognising a case; knowing when to question Unaided tests after the AI is taken away: students, programmers, support workers during outages Judgment at career distance is not yet measured.	Internal process, unwritten Routines; who knows what; whom to ask Organizational routines as the firm's memory; surgical trainees losing their share of the work No measurement of how new routines form with AI in use.

Note: Second Order framework, not a measurement. Each cell names what the studies in this issue measured there. Categories overlap, and no rate of movement between cells is measured.

Source: Polanyi; Nelson and Winter; Lave and Wenger; Beane; the output, learning and survey-data studies discussed in the text.

SO WHAT

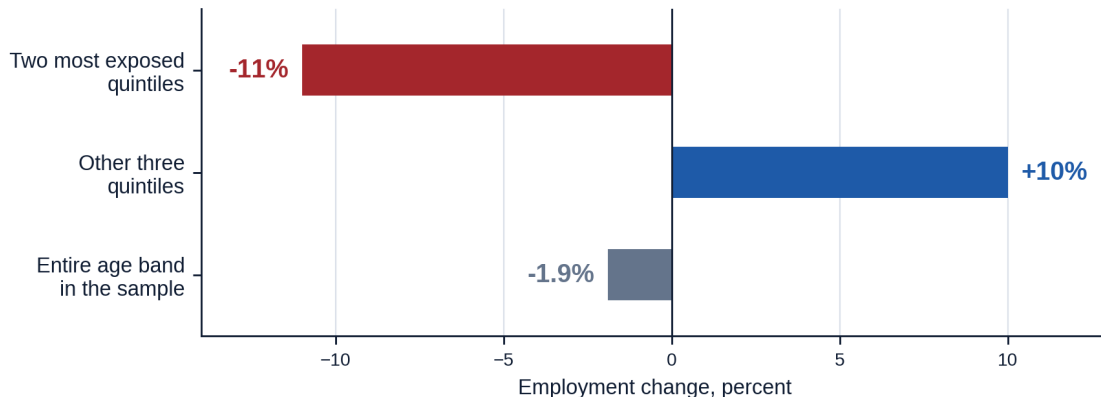
- Two distinctions sort every study in this issue: written down or learned by doing, held by the worker or by the firm.
- The four output experiments sit in the written-down cells. The two withdrawal trials sit in the worker's judgment cell. The withdrawal cell is where the evidence is thinnest.
- A firm that puts an AI in front of its juniors is deciding which parts of the work they still do by hand, and that decision fixes what they will know as seniors.

02 · EVIDENCE

EXHIBIT 2

Entry employment fell where the work was most exposed and rose where it was not

ADP payroll employment, ages 22 to 25, November 2022 to June 2026



Note: Stanford's rounded changes: about -11% in the two highest exposure quintiles and +10% in the other three; the whole age band fell 1.9%. The roughly 19% relative shortfall is $0.89 / 1.10 - 1$, not a 19% absolute decline. Changing age-band composition; this sample does not identify a national causal AI effect.

Source: Brynjolfsson, Chandar and Chen, Canaries, August 2026, p. 11; Second Order arithmetic from rounded inputs.

SO WHAT

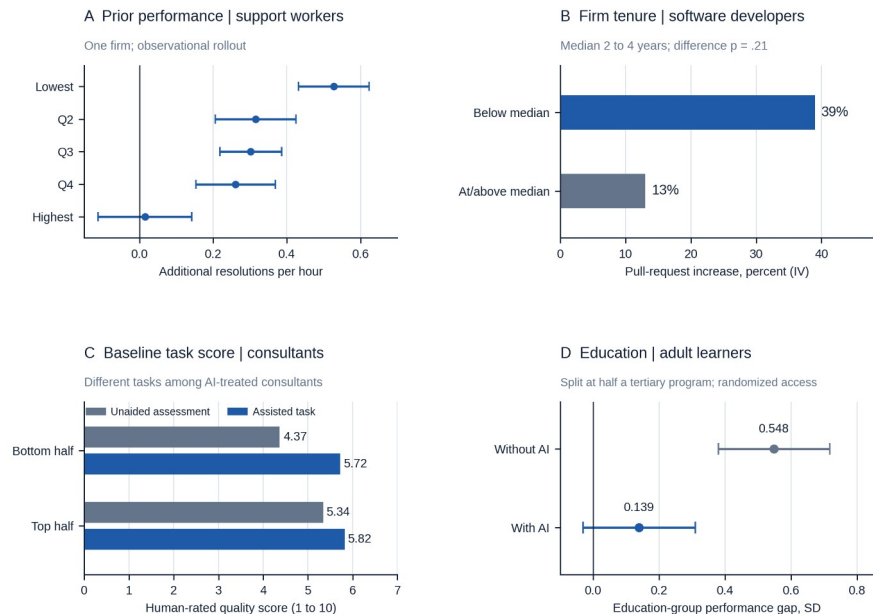
- By Stanford's count, employment of 22-to-25-year-olds in the two most AI-exposed occupation groups fell about 11 percent while the same age group in the other three grew about 10 percent.
- Issue 03 found those occupations growing across all ages with hiring frozen; the addendum found adopting firms hiring juniors. The jobs are growing and the youngest are not being brought in.
- Who is hiring is settled. What the hired junior learns once the AI does the written-down work is the separate measurement this issue takes.

03 · EVIDENCE

EXHIBIT 3

Four definitions of prior advantage, four gradients, no average

Four studies, four definitions of a newcomer; the panels share a direction, not a scale



Note: Different scales; no pooling. QJE is observational; Cui's tenure contrast is imprecise; Dell'Acqua follows AI users across different tasks; Cruces randomizes access. In the Dell'Acqua paper the upper-group percentage printed beside its figure does not follow from that figure's own levels, so the levels are plotted here and no percentage is. On a separate task outside the model's capabilities, AI reduced correctness 19 percentage points. Shown 95% intervals are house approximations from published standard errors.

Source: Published QJE Appendix Table A.VI; Cui et al., June 2025, Figure 6; published Dell'Acqua et al., Figure 4; Cruces et al., Table 1.

SO WHAT

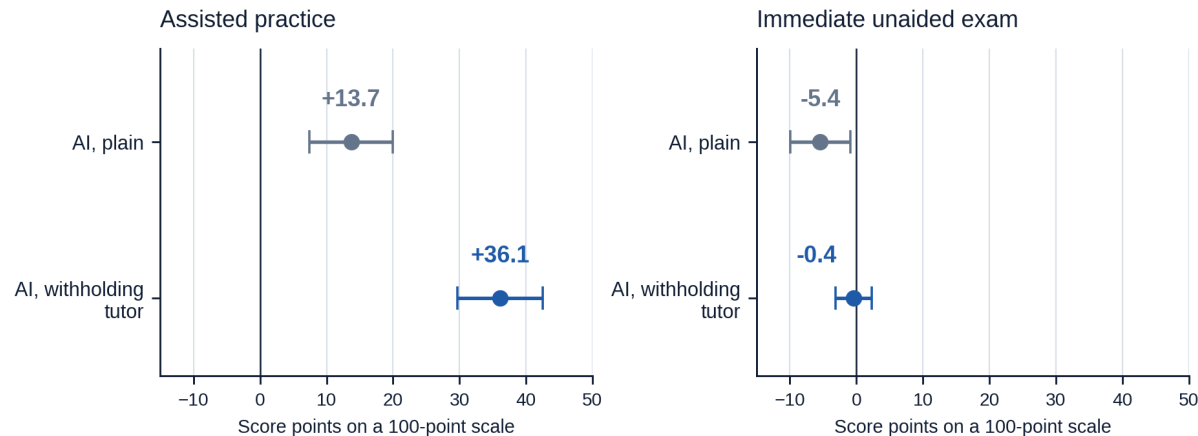
- Four experiments define a newcomer four ways: lowest prior performance, shortest tenure, lowest tested ability, least schooling. In every one, the people at the bottom improved most once they had the AI.
- The practice being handed over was the firm's own. The support-worker model was trained on the firm's top performers, and a new worker received their behavior as a suggested reply on day one.
- Output and speed in the first months used to show who was learning the job. The AI raises both directly, so they now grade the tool as much as the person.

04 · EVIDENCE

EXHIBIT 4

How the assistance was designed decided what remained when it was withdrawn

Same 825 students, same two AI tools. Left: scores during practice with the AI. Right: the exam afterwards with no AI.



Note: House reproduction of Bastani et al.'s main models: 825 students, 44 classrooms, four sessions at one Turkish school. Whiskers are classroom-clustered 95% intervals, using t with 43 degrees of freedom. The withholding tutor bundled hints, answer withholding and teachers' solutions and error guidance. Immediate unaided exams; no equivalence or delayed professional-learning result.

Source: Bastani et al., PNAS 2025, Table 1 and public author data; Second Order reproduction.

SO WHAT

- Practice scores rose under both AI tools. On the exam with no AI, the plain-AI group scored 5.4 points below students who never used one; the group whose tutor withheld answers scored the same as them.
- The tutor that avoided the loss came loaded with teachers' worked solutions and known-error guidance. A firm that buys the AI and skips the preparation has bought the version that cost 5.4 points.
- The four output experiments and the two withdrawal trials measure two different transfers. Output rose most for the newest; unaided ability, once the AI was gone, stayed flat or fell.

Five findings

1. Measure judgment separately

A junior's output no longer predicts what the junior knows. Test unaided work on purpose, because the first time you find out otherwise will be the day the AI is wrong on a case that matters.

2. Decide what juniors practice

The routine tasks the AI takes from an entry role were the tasks a junior learned on. The firm now has to choose what its juniors do by hand, because the AI no longer leaves them the routine work by default.

3. Buy the preparation, not just the tool

The AI tutor that did no harm came with teachers' solutions and known errors built in. That preparation is expert labor, and it is the part the purchase order does not include.

4. Your people know the model is built from them

Ninety-five percent of surveyed workers say they could withhold what they know; told their data trains an AI, refusals to hand over past work rose from a quarter to two fifths. Say what the data is for.

5. The apprenticeship no longer pays for itself

Junior work was billed, and a trained senior came free with it. The AI keeps the billing and drops the training. Teaching juniors is now a cost the firm funds deliberately.

WATCHLIST

What would change our mind

Senior review minutes: per piece of junior work. Review time is how a firm measures trust, and it falls as a junior earns it. If it holds or rises while AI lifts output, seniors are still supplying the judgment on every piece. Falling minutes are ambiguous on their own.

Promoted-cohort error rates: on work done without the AI, against earlier cohorts at the same stage.

Time to sign-off: in audit, law and medicine, where independence has a formal milestone. Shortening would run against this argument.

A longer learning trial: larger than 52 subjects, outside software, testing weeks or months after practice ends. If the deficit vanishes at that distance, the concern is smaller than the evidence here suggests.

The addendum's falsifier: whether the youth employment decline spreads from ages 22 to 25 into 26 to 30 in the next Stanford release.

SECOND-ORDER EFFECT · THE APPRENTICESHIP WAS FINANCED BY THE APPRENTICE. A junior's hours were billed like anyone else's, and the same hours produced a trained senior as a byproduct. When the AI does the work, billing continues and the byproduct is gone, so training becomes a separate cost. No source here measures whether cheaper teaching makes up the difference, or what the senior loses when the junior's questions stop.